

Problem_Set_3

Group 2

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1. If $e^{1.3}$ is approximated by Lagrangian interpolation from the values for $e^0 = 1, e^1 = 2.7183$ and $e^2 = 7.3891$, what are the minimum and maximum estimates for the error? Compare to the actual error. Answer:

2. Construct the divided-difference table from these data:

x	-0.2	0.3	0.7	-0.3	0.1
$f(x)$	1.23	2.34	-1.05	6.51	-0.06

Use the divided-difference table to interpolate for $f(0.4)$ Answer:

3. You have these values for x and $f(x)$:

x	-0.2	0.3	0.7	-0.3	0.1
$f(x)$	1.23	2.34	-1.05	6.51	-0.06

Find $f(0.5)$ from a cubic that starts from $x = 0.1$. Answer:

4. Given function $f(x) = 2x \cos 2x$. Use a central difference to compute $f'(2.0)$ and compare it using the $P'(x)$. Answer:

5. Use Trapezoid rule to estimate $\int_1^2 \frac{1}{x^2} dx$. Accurate within 0.001. Answer:

6. Use Simpson's rule to estimate $\int_1^2 \frac{1}{x^2} dx$. Accurate within 0.001. Answer: